

[ACJC/2022/Q4]

- 4 A library wants to store the details of its books in a relational database. The tables for the database have been created and stored in `Task4.db`.

There are two tables in the database:

- `books(bookID, title, price)`
- `copies(copyID, bookID)`

The library currently stores its information in two text files, `books_data.txt` and `copies_data.txt`. The library would like to reduce duplication of information by storing all the data into the relational database, `Task4.db`.

Each line in `books_data.txt` contains information regarding a book in the following format:

```
bookID,title,price
```

Each line in `copies_data.txt` contains information regarding a copy of a book in the library in the following format:

```
copyID,bookID,title,price
```

Task 4.1

Write a Python program to insert all information from the files into `Task4.db`.

Run the program.

[4]

Save your program code as

`TASK4_1_<your name>_<centre number>_<index number>.py`

Task 4.2

Write a Python program and the necessary files to create a web application. The web application offers the following menu options.

Insert a new book
Display all books

Save your program code as

`TASK4_2_<your name>_<centre number>_<index number>.py`

with any additional files/subfolders as needed in a folder named

`TASK4_2_<your name>_<centre number>_<index number>`

Run the web application and save the output of the program as

`TASK4_2_<your name>_<centre number>_<index number>.html`

[3]

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Task 4.3

The library would like to extend the web application to:

- receive bookID, title, price, and number of copies for a new book to be inserted into Task4.db;
- assign a copyID to each copy of the book;
- insert the data into the respective tables of Task4.db.

The copyID is a four-digit string starting with 0001. For example, if a book has three copies, they would have the copyIDs

- 0001
- 0002
- 0003

respectively.

The web page to receive the required information should be accessed from the corresponding menu option from **Task 4.1**. [6]

Test your program by inserting a book with the following details into Task4.db.

- bookID: 567420
- title: Plant farm
- price: 15.00
- number of copies: 4

[1]

Save your program code as

TASK4_3_<your name>_<centre number>_<index number>.py

With any additional files/subfolders as needed in a folder named

TASK4_3_<your name>_<centre number>_<index number>

Save your Python program as

TASK4_3_<your name>_<centre number>_<index number>.py

with any additional files / sub-folders as needed in a folder named

TASK4_3_<your name>_<centre number>_<index number>

Task 4.4

The library would like the web application to display information for all the books in a table.

Extend your web application to show the bookID, title, price, and the number of copies of that book in a table. The results should be shown in alphabetical order by book title.

The resulting web page should be accessed from the corresponding menu option from **Task 4.1**. [6]

Save your program code as

TASK4_4_<your name>_<centre number>_<index number>.py

with any additional files/subfolders as needed in a folder named

TASK4_4_<your name>_<centre number>_<index number>

Run the web application and save the output of the program as

TASK4_4_<your name>_<centre number>_<index number>.html

[2]